for some family of unitary operators $\hat{U}(t)$. If the Hamiltonian is independent of time then $\hat{U}(t) = e^{-i\hat{H}t}$ (in units where $\hbar = 1$).

Because of the unitarity of time evolution, all the basic operations in quantum computing and information processing (except measurements) are described by unitary operators.

2.1.3 Measurements and observables

The *measurement postulate* dictates how measurements on physical systems are modelled in quantum mechanics. We can describe measurements in two seemingly different but ultimately equivalent ways.

POVMs

A general measurement in quantum mechanics is associated with a set of measurement operators $\{\hat{M}_m\}$ where m labels the possible measurement outcomes. For a system in a normalised state $|\psi\rangle$, the probability for the outcome m to occur is

$$p(m) = \langle\psi|\hat{M}_m^\dagger \hat{M}_m|\psi\rangle. \tag{2.7}$$

This is known as the *Born rule*. Because this expression is the norm of the vector $\hat{M}_m|\psi\rangle$ (or equivalently, because the operator $\hat{E}_m = \hat{M}_m^\dagger \hat{M}_m$ is positive semidefinite), the probabilities are necessarily non-negative, $p(m) \geq 0$. In addition, the measurement operators must satisfy the completeness relation

$$\sum_m \hat{M}_m^\dagger \hat{M}_m = \mathbb{1}, \tag{2.8}$$

which ensures that the probabilities sum to one,

$$\sum_m p(m) = \sum_m \langle\psi|\hat{M}_m^\dagger \hat{M}_m|\psi\rangle = \langle\psi|(\sum_m \hat{M}_m^\dagger \hat{M}_m)|\psi\rangle = 1, \tag{2.9}$$

for any normalised state $|\psi\rangle$. The (normalised) state of the system immediately after a measurement with outcome m is given by

$$\frac{\hat{M}_m|\psi\rangle}{\sqrt{\langle\psi|\hat{M}_m^\dagger \hat{M}_m|\psi\rangle}} = \frac{1}{\sqrt{p(m)}}\hat{M}_m|\psi\rangle. \tag{2.10}$$

The set of positive operators $\{\hat{E}_m\}$ is called a *positive-operator-valued measure* (POVM). Any set of operators satisfying

$$\hat{E}_m^\dagger = \hat{E}_m \qquad \text{(Hermiticity)}, \tag{2.11}$$

$$\hat{E}_m \geq 0 \qquad \text{(positivity)}, \tag{2.12}$$

$$\sum_m \hat{E}_m = \mathbb{1} \qquad \text{(completeness)}, \tag{2.13}$$

form a POVM and the operators $\hat{E}_m$ are referred to as POVM elements. POVMs provide the most general description of measurements in quantum mechanics. The outcome probabilities are given by $p(m) = \langle\psi|\hat{E}_m|\psi\rangle$. Note that the decomposition of a POVM into measurement operators, $\hat{E}_m = \hat{M}_m^\dagger \hat{M}_m$,

is generally not unique. Hence, only specifying the POVM does not fix the post-measurement state. However, when the post-measurement state is not of interest, it is often convenient to work just with POVMs. In particular, the form of the conditions (2.11)-(2.13) imply that optimisation over arbitrary measurements can be realised using powerful methods of semidefinite programming (a type of convex optimisation). Only the positivity and completeness constraints need to be imposed in practice, because non-negativity implies Hermiticity for operators on a complex Hilbert space.

Projective measurements and observables

An important class of POVMs are those for which the POVM elements are projectors. That is, POVMs with elements $\{\hat{P}_m\}$ such that $\hat{P}_m\hat{P}_n = 0$ for $n \neq m$ and $\hat{P}_n^2 = \hat{P}_n$. These are known as projective measurements and are associated with observables.

An *observable* is a measurable physical property of a system. In quantum mechanics, an observable is any Hermitian operator $\hat{M}$. A Hermitian operator has a spectral decomposition and can be written

$$\hat{M} = \sum_m m\hat{P}_m \tag{2.14}$$

where each m is an eigenvalue of $\hat{M}$ and $\hat{P}_m$ is the orthogonal projector onto the corresponding eigenspace spanned by the eigenvectors with eigenvalue m. The Hermiticity of $\hat{M}$ implies that the eigenvalues are real. An observable thus defines a projective measurement with outcomes given by its eigenvalues (and any projective measurement with real outcomes defines an observable via Eq. (2.14)). For projective measurements, since $\hat{P}_m^\dagger\hat{P}_m = \hat{P}_m^2 = \hat{P}_m$, the POVM elements are themselves measurement operators $\hat{M}_m = \hat{P}_m$ and the (unnormalised) post-measurement state upon obtaining outcome m is $\hat{P}_m|\psi\rangle$. If an eigenvalue m is non-degenerate then $\hat{P}_m$ is the projector onto the corresponding eigenvector, $\hat{P}_m = |m\rangle\langle m|$, and the post-measurement state is then simply $|m\rangle$.

Equivalence of POVMs and observables

From the above, it might seem that POVMs are more general than observables and projective measurements, and POVMs are also sometimes referred to as 'generalised measurements'. In particular, a projective measurement on a d-dimensional system can have at most d outcomes, because projectors corresponding to distinct outcomes must be orthogonal. General POVMs have no such restriction and can have any number of outcomes.

However, any POVM can in fact be realised by letting the system interact unitarily with another, auxiliary system of sufficiently large dimension and then performing a projective measurement on the joint system. In this sense, POVMs and observables are mathematically equivalent. Projective measurements (together with unitary evolution and composition of systems) are sufficient to describe every measurement quantum mechanics can do in principle. In practice, though, it is often much simpler to work with POVMs.

Example 2.1: State discrimination POVM

As an example of how POVMs can be useful, consider the task of discriminating between two non-orthogonal states, $|\psi_0\rangle$ and $|\psi_1\rangle$, of a two-dimensional system.

A measurement device receives $|\psi_x\rangle$, with $x \in \{0, 1\}$ and outputs an outcome m. The goal is to identify the state, i.e. to output $m = x$. The behaviour of the device is characterised by the conditional probabilities $p(m|x)$. The probabilities for successfully identifying $|\psi_0\rangle$ and $|\psi_1\rangle$ are $p(0|0)$ and $p(1|1)$, respectively, while $p(1|0)$ and $p(0|1)$ are the probabilities for erroneously identifying $|\psi_0\rangle$ as $|\psi_1\rangle$ and vice versa.

If the states are not orthogonal, then there is no projective measurement on the system which can discriminate them without error. This can be seen as follows. Let $\delta = |\langle\psi_0|\psi_1\rangle| > 0$ denote the overlap. Since the measurement is projective, it can have at most two outcomes. The probabilities are $p(m|x) = \langle\psi_x|\hat{P}_m|\psi_x\rangle$, where $\hat{P}_m$ is the projector corresponding to outcome m. If the device makes no errors in identifying the first state, $p(1|0) = 0$, then we must have $p(0|0) = 1$, since the probabilities must sum to one. But this implies that $\hat{P}_0 = |\psi_0\rangle\langle\psi_0|$ and therefore $p(0|1) = |\langle\psi_0|\psi_1\rangle|^2 = \delta^2 > 0$. Similarly, if $p(0|1) = 0$ then $p(1|0)$ must be nonzero. Thus, for any projective measurement, either one or the other state (or both) cannot be unambiguously identified.

With a POVM, on the other hand, we can ensure that both states are unambiguously identified, by allowing for a third, inconclusive outcome denoted $\varnothing$. In particular, define a three-outcome POVM with elements

$$\hat{E}_0 = \frac{1}{1+\delta}|\psi_1^\perp\rangle\langle\psi_1^\perp|, \tag{2.15}$$

$$\hat{E}_1 = \frac{1}{1+\delta}|\psi_0^\perp\rangle\langle\psi_0^\perp|, \tag{2.16}$$

$$\hat{E}_\varnothing = \mathbb{1} - \hat{E}_0 - \hat{E}_1, \tag{2.17}$$

where $|\psi_x^\perp\rangle$ is the state orthogonal to $|\psi_x\rangle$. The set $\{\hat{E}_0, \hat{E}_1, \hat{E}_\varnothing\}$ is complete by definition and clearly $\hat{E}_0, \hat{E}_1 \geq 0$. It is not difficult to show that also $\hat{E}_\varnothing \geq 0$. Hence, this is a valid POVM. The error probabilities vanish, $p(1|0) = \langle\psi_0|\hat{E}_1|\psi_0\rangle \propto |\langle\psi_0|\psi_0^\perp\rangle|^2 = 0$ and similarly $p(0|1) = 0$. The success probability for identifying the first state is

$$p(0|0) = \frac{|\langle\psi_0|\psi_0^\perp\rangle|^2}{1+\delta} = \frac{1 - |\langle\psi_0|\psi_1\rangle|^2}{1+\delta} = \frac{1-\delta^2}{1+\delta} = 1 - \delta, \tag{2.18}$$

and similarly $p(1|1) = 1-\delta$. Thus, this POVM unambiguously identifies both states – when it outputs 1 or 2 this answer is always correct – but at the price of sometimes providing an inconclusive result. The probability of an inconclusive outcome is

$$p(\varnothing|x) = 1 - p(0|x) - p(1|x) = \delta. \tag{2.19}$$